

University of Information Technology and Sciences (UITS)

Department of Computer Science and Engineering

COURSE: Software Project Design and Development (CSE 416)

PROJECT PROPOSAL 2.0

Project Title: CampusCare

Centralized University Complaint Management System

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1 Objectives

1.1 Primary Objective

To design, develop, and deploy a responsive campus facility management platform that centralizes complaint submission, automated Service Level Agreement (SLA) deadline tracking, role-based dispatching, rate-limited submission APIs, and audit logging within a unified system.

1.2 Quantifiable and Operational Objectives

1. **Resolution Acceleration:** Reduce average complaint resolution lifecycle time across campus facility operations by 40%.
2. **Centralized Tracking:** Achieve 100% centralized tracking of submitted service requests through an authenticated single submission pipeline.
3. **SLA Compliance:** Maintain institutional SLA compliance above 85% across all complaint priority levels (`Critical: 4h`, `High: 24h`, `Medium: 72h`, `Low: 168h`).
4. **Database Query Performance:** Maintain database response latencies below 500ms for administrative reporting queries on datasets up to 10,000 records.
5. **Audit Traceability:** Maintain 100% audit coverage for ticket status transitions and staff assignment updates stored in an immutable audit ledger (`audit_logs`).

2 Scope

2.1 In-Scope (Minimum Viable Product - MVP)

The scope for the initial system release encompasses:

- **Authentication and Security:** Email/password authentication via `@supabase/ssr`, `HttpOnly` session cookie management, Next.js Edge proxy middleware for role-based route protection (`Student`, `Staff`, `Admin`), and Upstash Redis sliding-window rate limiting (`@upstash/ratelimit`) on submission endpoints.
- **Student Module:** Dashboard with metric cards, complaint submission form with custom animated dropdown pickers and photo proof uploads, human-readable Complaint ID generation (`CMP-YYYY-XXXX`), real-time ticket timeline tracking, and resolution confirmation/rejection workflow.
- **Staff Module:** Departmental ticket queues, visual SLA deadline warning indicators,

state transition controls (`Assigned` → `In Progress` → `Resolved`), internal progress note logs, manual SLA target date overrides with reason notes, and completion photo proof uploads.

- **Admin Module:** Global master ticket table with multi-parameter filtering using `@tanstack/react-table`, ticket reassignment and priority override controls, user role promotion and department assignment desk, category/department taxonomy management, and operational analytics charts.
- **Database and Storage Infrastructure:** PostgreSQL relational schema with Row Level Security (RLS) policies utilizing `SECURITY DEFINER` helper functions to prevent policy recursion, foreign key constraints, automated ticket sequence triggers, and an immutable audit log table (`audit_logs`).

2.2 Out-of-Scope (Future Iterations - Version 2.0)

The following capabilities are excluded from the initial release and reserved for future project phases:

- Automated AI complaint classification and duplicate ticket detection.
- Real-time WebSocket push notifications and external transactional email dispatching (e.g., Resend / SendGrid API).
- Native mobile applications for iOS/Android operational staff (React Native / Expo).
- Multi-campus organizational hierarchy support across geographically separated university branches.

3 Feasibility Study

3.1 Technical Feasibility

The system leverages Next.js 16 (App Router), React 19, TypeScript, Tailwind CSS, `@tanstack/react-table`, and Supabase (PostgreSQL). This stack provides standard SSR support, strong type safety, and clean separation of client and server logic. Supabase Row Level Security (RLS) combined with `SECURITY DEFINER` SQL helper functions enforces strict role-based data isolation at the database layer. Upstash Redis handles request rate limiting to protect public endpoints against abuse.

3.2 Operational Feasibility

The UI is built using the Caring Campus Green design system with responsive layouts supporting viewports down to 320px width and high-contrast text meeting WCAG standards. The system reflects standard university facility workflows (`Student reports` → `System/Admin assigns` → `Staff resolves` → `Student verifies`), requiring minimal training for campus maintenance personnel and students.

3.3 Economic Feasibility

Next.js, React, Tailwind CSS, and PostgreSQL incur zero software licensing fees. Hosting on Vercel and Supabase keeps operational overhead low, starting on free dev tiers and scaling to modest Pro tiers ($45/month total) for production workloads, significantly reducing manual administrative labor costs.

4 Work Breakdown Structure (WBS)

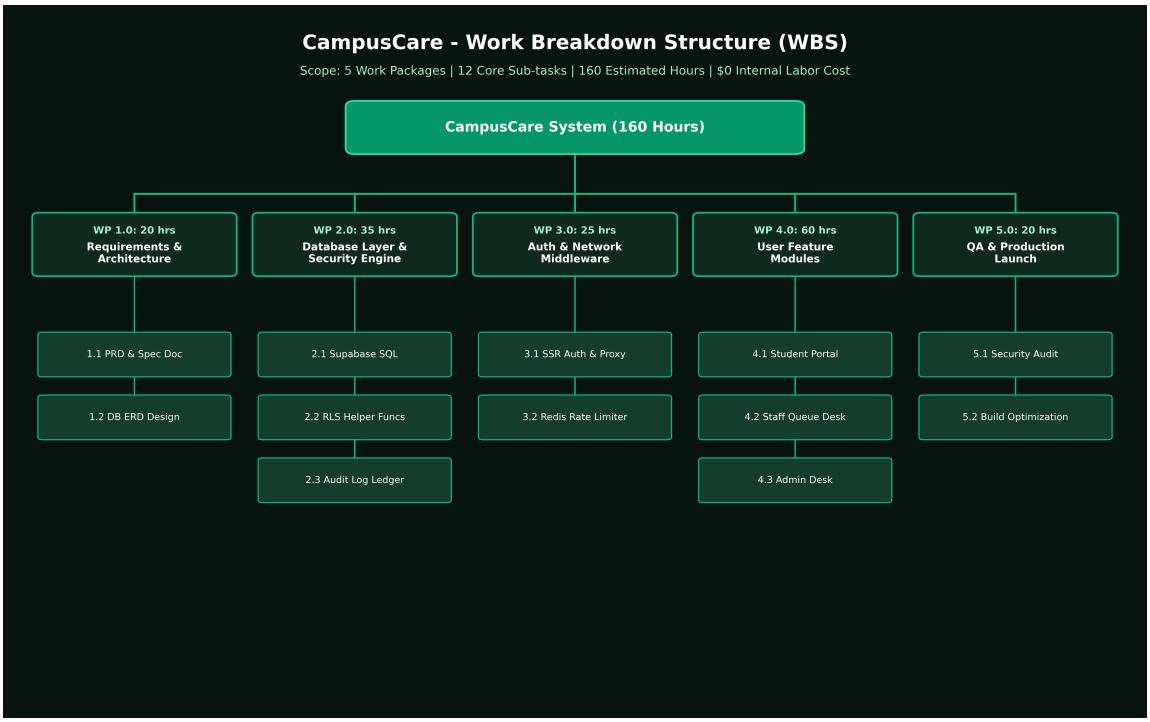


Figure 1: CampusCare Work Breakdown Structure (WBS) Tree Diagram.

Work Package	Title and Deliverables	Labor (Hours)	Cost (USD)
WP 1.0	Requirements and Architecture (PRD, Spec, ERD)	20 hrs	$0.00
WP 2.0	Database and Security (PostgreSQL, RLS, Audit Log)	35 hrs	$0.00
WP 3.0	Auth and Middleware (`@supabase/ssr`, Proxy, Redis)	25 hrs	$0.00
WP 4.0	Core Modules (Student, Staff, Admin Desks)	60 hrs	$0.00
WP 5.0	QA and Launch (Security Audit, WCAG, Build Optimization)	20 hrs	$0.00
Total Project	**5 Work Packages (12 Sub-tasks)**	**160 Hours**	**$0.00**

Table 1: Work Package Task and Labor Costing Breakdown.

5 Gantt Chart and Project Timeline

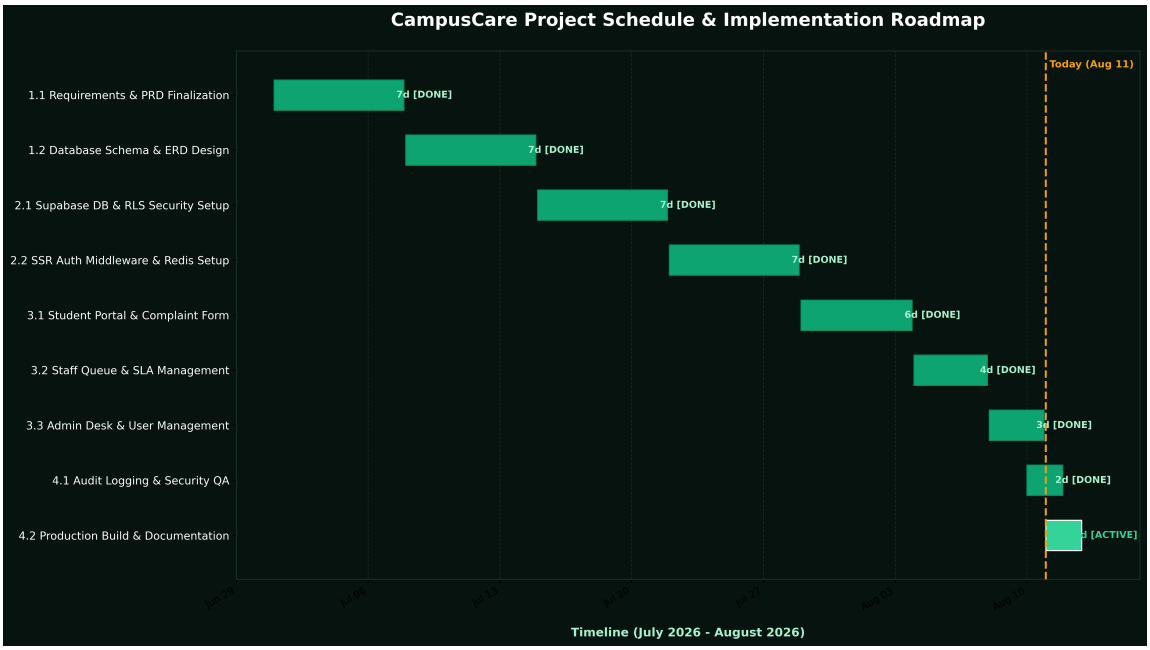


Figure 2: CampusCare Project Implementation Schedule and Gantt Chart.

6 Budget and Cost Estimation

Category	Component / Resource	Quantity	Annual Cost (USD)
Development	Internal Engineering Team	1 Team	$0.00
Application Hosting	Vercel Hosting (Pro Tier)	12 Months	$240.00
Database and Auth	Supabase Managed PostgreSQL (Pro Tier)	12 Months	$300.00
Rate Limiting	Upstash Redis Cloud	12 Months	$120.00
Domain and Network	Campus Subdomain (`campuscare.university.edu`)	1 Domain	$0.00
SSL Certificate	Vercel / Let's Encrypt Automated SSL	1 Certificate	$0.00
Buffer / Storage	Bandwidth and Storage Contingency Buffer	1	$100.00
Total Annual Cost			**$760.00**

Table 2: Annualized Cloud Infrastructure and Operational Budget Breakdown.